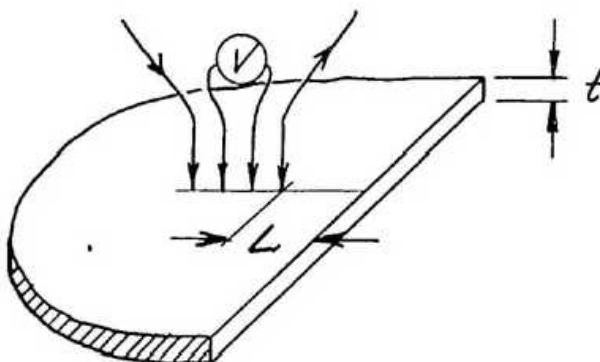


E.3) Probe Array Perpendicular to Edge, Thin Sample.

$$t \leq 2s$$

When the sample is thin, it is convenient to express the resistivity as follows :

$$\rho = \frac{G V}{I}, \quad G = \frac{\pi}{\ln 2} \cdot t \cdot D_3\left(\frac{L}{s}\right) \cdot F_2\left(\frac{t}{s}, \frac{L}{s}\right) \quad (13)$$

where

$\frac{\pi}{\ln 2} \cdot t = 4.5324 \cdot t$ is the geometric factor for an infinite slice of thickness $t \ll s$,

$$D_3\left(\frac{L}{s}\right) = \frac{1}{1 + \frac{1}{2 \ln 2} \ln \frac{(\frac{L}{s} + 2)(\frac{L}{s} + 1)}{(\frac{L}{s} + \frac{5}{2})(\frac{L}{s} + \frac{1}{2})}} \quad (14)$$

is the additional correction to apply when measuring at a distance L from the edge.

$$D_3\left(\frac{L}{s}\right) \rightarrow 1 \text{ as } \frac{L}{s} \rightarrow \infty$$

$F_2\left(\frac{t}{s}, \frac{L}{s}\right)$ deviates from unity when the thickness t of the slice becomes comparable to the probe distance s .